Onshape Workflow Manual

Wind Turbine Design Lab exports an Onshape package so students can rebuild a selected blade in CAD before creating an STL for 3D printing. This manual walks through the workflow slowly, assuming the user is new to Onshape.

This export is not a finished STL. It is a CAD reference package: it gives the blade table, 2D outlines, and airfoil section profiles. Students still need to build the 3D part in Onshape, check the hub connection, and export STL from Onshape.

1. Files in the downloaded ZIP

Click **Download Onshape package** in the app, then unzip the file. You should see these files:

File	Use
`blade_geometry.csv`	Main table: section, position, chord, twist, airfoil, and role.
`airfoil_sections.csv`	Airfoil metadata and recommended Reynolds-number ranges.
`blade_planform.dxf`	Top-view blade outline in centimetres. Use it as an outline check.
`section_profiles/section_XX_*.dxf`	One ready-to-insert DXF per blade section. Use these for the main workflow.
`section_profiles/README.md`	List of the individual section DXF files and their chord/twist values.
`section_profiles.dxf`	Combined profile overview. Use it only as a visual reference.
`design_metadata.json`	Simulator settings saved with the export.
`onshape_build_guide.md`	Short build notes included inside the export.

For the first Onshape build, keep `blade_geometry.csv` open beside Onshape. It is the map for where every airfoil section belongs.

2. Read the geometry table first

Open `blade_geometry.csv`. The important columns are:

Column	Meaning
`Section`	Root-to-tip section number.
`Position (cm)`	Distance from the turbine center along the blade radius.
`Chord (cm)`	Airfoil width from leading edge to trailing edge.
`Twist (deg)`	Rotation angle for that section.
`Airfoil`	Airfoil profile assigned to that section.
`Role`	Plain-language reason for using that airfoil there.

Example:

```
Section,Position (cm),Chord (cm),Twist (deg),Airfoil
1 (Root),6.0,9.0,20.0,NACA 4418
2,14.0,7.5,15.0,SG6040
3,24.0,5.8,10.0,SG6043
4,34.0,4.2,6.0,SG6043
5,43.0,3.0,2.0,NACA 2412
6 (Tip),50.0,1.8,0.0,E387
```

The table positions are measured from the rotor center. In CAD it is usually easier to start the blade at the first section. If section 1 is at 6 cm and section 6 is at 50 cm, then the CAD station offsets from section 1 are:

Section	Table position	CAD offset from section 1
1	6 cm	0 cm
2	14 cm	8 cm
3	24 cm	18 cm
4	34 cm	28 cm
5	43 cm	37 cm
6	50 cm	44 cm

Formula:

```
CAD offset = section position - root section position
```

3. Create a new Onshape document

1. Open Onshape.
2. Create a new Document.
3. Open a Part Studio.
4. Use centimetres as the working unit when importing DXF files.

If a sketch imports much too large or much too small, the unit was probably wrong. Delete the import and insert it again with centimetres.

4. Import the top-view planform

The planform is a top-view blade outline. It is useful as a visual guide, but it is not the main Loft input.

1. Start a Sketch on the Top plane.
2. Insert or import `blade_planform.dxf`.
3. Choose centimetres if Onshape asks for units.
4. Keep this sketch visible as a reference.

Use this file to check whether the final blade outline looks similar to the simulator's top-view design.

5. Create section planes

Create one plane for each blade section. The planes should be perpendicular to the blade length direction.

For the example table above:

Plane name	Offset from first/root section
`S1 Root`	0 cm
`S2`	8 cm

Plane name	Offset from first/root section
`S3`	18 cm
`S4`	28 cm
`S5`	37 cm
`S6 Tip`	44 cm

The actual numbers must come from your own `blade_geometry.csv`.

What S1 Root means in Onshape

When this manual says **use the Right plane as S1 Root**, it means:

```
Do not create a new plane for section 1.
The existing Right plane is section 1 / root.
```

In the Part Studio feature tree, expand **Default geometry**. You should see:

```
Origin
Top
Front
Right
```

Use `Right` as the first root section plane. Later, when you sketch the first airfoil profile, create that sketch directly on `Right`.

Create the offset planes for S2 to S6

1. If a failed Plane dialog is open, click the red X to cancel it.
2. Expand **Default geometry** in the left feature tree.
3. Click the **Plane** tool.
4. Set the plane type to **Offset**.
5. Click the `Entities` box.
6. Select the existing `Right` plane from the feature tree or from the graphics area.
7. Change the distance from the default value, often `1 in`, to the required centimetre value.
8. For S2, type `8 cm`.
9. Click the green check mark.

Repeat the same process using `Right` as the base plane every time:

New plane	Base plane to select	Offset distance
`S2`	`Right`	`8 cm`
`S3`	`Right`	`18 cm`
`S4`	`Right`	`28 cm`
`S5`	`Right`	`37 cm`
`S6 Tip`	`Right`	`44 cm`

Important: do not leave the distance as `1 in`. Type the centimetre unit directly, such as `8 cm`, `18 cm`, or `44 cm`.

If the new plane appears on the wrong side, use **Flip normal** in the Plane dialog.

After creating each plane, rename it immediately. Clear names make Loft much easier.

Common plane creation error

If Onshape shows this message:

```
Offset plane requires a plane, circle, ellipse, or arc to offset from.
```

The `Entities` box is still empty. You clicked the Plane tool, but you did not select a base plane yet.

Fix:

```
Plane tool
-> Offset
-> click Entities
-> click Right plane
-> type 8 cm
-> green check
```

For this workflow:

```
Right plane = S1 Root
Offset 8 cm from Right = S2
Offset 18 cm from Right = S3
Offset 28 cm from Right = S4
Offset 37 cm from Right = S5
Offset 44 cm from Right = S6 Tip
```

6. Place airfoil profiles on the planes

Use the individual files inside the `section_profiles/` folder. The folder contains one ready-to-insert DXF per blade section. Each file contains one airfoil only, scaled to the correct chord, centred near the sketch origin, and pre-rotated by the section twist angle.

For example:

Plane	Insert this DXF
`S1 Root`	`section_profiles/section_01_root_NACA4418.dxf`
`S2`	`section_profiles/section_02_SG6040.dxf`
`S3`	`section_profiles/section_03_SG6043.dxf`
`S4`	`section_profiles/section_04_SG6043.dxf`
`S5`	`section_profiles/section_05_NACA2412.dxf`
`S6 Tip`	`section_profiles/section_06_tip_E387.dxf`

These filenames are examples from one preset. Your package may use different airfoil names, so always check `section_profiles/README.md` or `blade_geometry.csv`.

This is the beginner workflow:

1. Create a Sketch on plane `S1 Root`.
2. Insert `section_profiles/section_01_root_NACA4418.dxf`.
3. Choose centimetres if Onshape asks for units.
4. Confirm the sketch.
5. Create a Sketch on plane `S2`.
6. Insert the section 2 DXF file.
7. Repeat from root to tip.

With the individual section DXFs, you do not need to delete the other sections and you usually do not need to manually rotate twist. The individual files are already prepared for that section.

Do not use `section_profiles.dxf` as the main beginner workflow. That combined file is kept only as an overview/reference drawing because it contains all sections together.

7. Apply twist correctly

Twist is the local rotation of each airfoil section.

The individual `section_profiles/section_XX_*.dxf` files are already pre-rotated by the twist angle in `blade_geometry.csv`. For the beginner workflow, insert the individual section file and do not rotate it again.

For each sketch in the beginner workflow:

Section	Twist handling
Section 1	Already applied in the section 1 DXF file
Section 2	Already applied in the section 2 DXF file
Section 3	Already applied in the section 3 DXF file
Section 4	Already applied in the section 4 DXF file
Section 5	Already applied in the section 5 DXF file
Section 6	Already applied in the section 6 DXF file

Use the twist values in `blade_geometry.csv` for checking only. Do not rotate the individual DXF files again unless you intentionally edit the CAD model.

Keep the rotation direction consistent. If the Loft appears twisted backward, undo, reverse the rotation direction, and try again.

8. Align the profiles before Loft

Loft works best when all profiles share a consistent reference point.

Pick one alignment method and use it for every section:

- align each profile by chord midpoint; or
- align each profile by leading edge; or
- align each profile by trailing edge.

For classroom blades, chord midpoint alignment is usually easiest. Leading-edge alignment can also work, but do not mix alignment methods section by section.

Before Loft, check:

- every sketch has exactly one closed airfoil profile;
- profiles are ordered from root to tip;
- the same side of the airfoil faces the same direction on every plane;
- twist is applied in the intended direction.

9. Create the 3D blade with Loft

1. Start the Loft feature.
2. Choose **Solid** if each profile is a closed region.
3. Select the section sketches in order:

- section 1 root profile; - section 2 profile; - section 3 profile; - section 4 profile; - section 5 profile; - section 6 tip profile.

4. Preview the shape.
5. If the blade twists strangely, cancel and check profile alignment and order.
6. Accept the Loft.

If Solid Loft fails, try Surface Loft first. If Surface Loft works but Solid Loft does not, at least one profile may not be a clean closed region.

10. Understand what `blade_planform.dxf` is for

`blade_planform.dxf` is not required for Loft. The Loft comes from the individual airfoil section DXFs.

Use `blade_planform.dxf` as a top-view reference:

1. Open the `blade_planform.dxf` tab in Onshape to see the intended blade outline.
2. Return to `Part Studio 1`.
3. Rotate the view toward Top view.
4. Compare the 3D Lofted blade to the planform outline.

If the Lofted blade is much wider, shorter, longer, or shifted sideways compared with the planform, check the imported DXF units and the profile alignment before continuing.

11. Add a beginner hub

The simulator exports blade geometry, not the final hub. You must add a real mount that matches the generator shaft and competition rules.

Important: Do not create the hub sketch on a Mate connector. Mate connectors are useful later for assemblies, but they confuse the beginner workflow. Create the hub sketch on a real plane near the root, usually `S1 Root`, `Right`, `Front`, or `Top`, depending on which view gives a clean circle at the rotor center.

Beginner hub dimensions

These are safe starting dimensions for a classroom test part. Change them to match the real generator shaft.

Feature	Suggested value	Notes
Hub outside diameter	`8 cm` to `12 cm`	Large enough to grip the blade roots.
Shaft hole diameter	measured shaft + clearance	Example: `0.8 cm` if the shaft is near 8 mm.
Hub thickness	`2 cm` to `4 cm`	Thicker hubs are easier to print and stronger.
Root overlap	`0.5 cm` to `2 cm`	Let blade roots enter the hub area.

Create the hub sketch

1. Cancel any open sketch that is on `Mate connector`.
2. If a blank `Sketch 7` was created on `Mate connector`, delete it.
3. Select a real plane near the root, such as `S1 Root` or `Right`.
4. Right-click the plane and choose **New sketch**.
5. Click **View normal to** so the sketch is flat on the screen.
6. Use the Circle tool to draw the outside hub circle.
7. Use the Circle tool again to draw the shaft hole circle in the center.
8. Add dimensions for both circles.
9. Confirm the sketch with the green check mark.

If the circle is not centered on the intended rotor axis, move it before extruding. The future 3-blade pattern will rotate around this hub center.

Extrude the hub

1. Start **Extrude**.
2. Select the ring-shaped area between the outside circle and shaft hole.
3. Operation: choose **Add** if it touches the blade, or **New** if it does not.
4. Distance: start with `2 cm` or `3 cm`.
5. If it extrudes in the wrong direction, flip the arrow.
6. Accept the feature.

If the hub is a separate part, use **Boolean > Union** later to combine the hub and blade bodies.

12. Reinforce and connect the blade root

The blade root is where most classroom prints fail. Before making three blades, make sure one blade is physically connected to the hub.

Use one of these beginner methods:

- Make the hub cylinder overlap the root of the blade slightly.
- Add a small rectangular or rounded root bridge from the blade root into the hub.
- Use **Boolean > Union** to merge the blade and hub if they are separate parts.
- Add a small fillet at the root if Onshape can calculate it cleanly.

Do not leave the blade touching the hub at only a thin edge. It may look correct on screen but break during wind-tunnel testing.

13. Create three blades with Circular Pattern

After one blade and hub connection look correct, make the three-blade rotor.

Recommended beginner settings:

```
Tool: Circular Pattern
Pattern type: `Part pattern`
Quantity: `3`
Angle: `360 deg`
Equal spacing: on
Axis: hub center axis
```

Steps:

1. Start **Circular Pattern**.
2. Set Pattern type: `Part pattern`.
3. Select the blade part. If the blade and hub are already one part, select the

part carefully and check the preview. Patterning the hub can create duplicate hubs, so it is often easier to pattern only the blade first.

4. Click the Axis selection box.
5. Select the cylindrical face of the hub or the circular edge of the shaft hole.
6. Set Quantity: 3.
7. Set Angle: `360 deg`.
8. Confirm that the preview shows blades 120 degrees apart.
9. Accept the feature.

If the pattern rotates around the wrong point, undo and choose the hub cylinder or shaft-hole circular edge as the axis again.

If you see an `S7 Tip` plane, remember: S7 Tip plane is not an airfoil profile. Only select actual blade or hub parts for Circular Pattern, not the blue section planes.

14. Combine the final rotor parts

The final export should usually be one clean rotor part.

1. If the patterned blades and hub are separate parts, start **Boolean**.
2. Choose **Union**.
3. Select the hub and all three blade bodies.
4. Accept the feature.
5. Check the Parts list. Ideally there should be one final rotor part.

If Boolean Union fails, one or more parts may not be touching. Increase the hub overlap or add a root bridge, then try Union again.

15. Check diameter and printability

Before exporting STL, check the model against the competition rule and printer limits.

For a 1 m diameter rule:

```
Maximum radius from rotor center to farthest tip = 50 cm
Maximum full diameter tip-to-tip = 100 cm
```

Use Onshape's Measure tool:

1. Measure from the hub center to the farthest blade tip.
2. Confirm the radius is not greater than `50 cm`.
3. Measure the shaft hole diameter.
4. Check the hub thickness.
5. Rotate the model and inspect root connections from both sides.

Printability checklist:

- no paper-thin root connection;
- no accidental duplicate hub bodies;
- no floating blade that is not joined to the hub;
- shaft hole is not smaller than the real shaft;
- blade tips are not outside the competition diameter;
- there is enough clearance for the wind-tunnel frame.

16. Export STL from Onshape

When the CAD model is ready:

1. In the left Parts list, right-click the final rotor part.
2. Choose **Export**.
3. Export format: `STL`.
4. STL format: choose `Binary STL` unless your slicer requires ASCII.
5. Units: choose millimetres if your slicer expects millimetres.
6. Resolution: choose Fine for a smoother airfoil, or Medium for a smaller file.
7. Confirm the export and download the `.stl` file.

Open the STL in your slicer before printing. Check the real dimensions again in the slicer because unit mistakes are common.

17. Slicer checks before printing

A successful STL export does not guarantee the blade is strong enough.

Before printing, check:

- scale is correct;
- rotor diameter is still within the rule;
- hub hole size matches the generator shaft;
- print orientation does not make the root weak;
- support material will not damage the airfoil surface too much;
- infill and wall count are high enough near the hub;
- the blade root has enough material to survive startup torque.

For a first classroom prototype, print only one small section or one blade first if time and budget allow. Then print the full three-blade rotor.

18. Troubleshooting

Problem	Likely cause	Fix
DXF imports at the wrong size	Wrong import unit	Reimport using centimetres.
Loft fails	Profile is not closed or contains extra curves	Clean the sketch; leave one closed profile per section.
Loft twists into a knot	Profiles selected out of order or aligned inconsistently	Select root to tip; align all profiles by the same reference point.
Blade looks mirrored	Profile orientation is flipped on one or more planes	Flip or rotate the affected sketch.
Blade outline does not match planform	Profiles are shifted sideways	Use `blade_planform.dxf` as a top-view reference and realign profiles.
Final part is too large	Rotor diameter or unit conversion issue	Check radius and units before exporting STL.
Root breaks easily	Hub connection too thin	Add root reinforcement, fillets, or a thicker hub mount.
Cannot select the last plane	The last plane is only a reference plane	Select sketch faces or parts, not blue planes.
Circular Pattern duplicates the hub	Hub and blade were already one part	Pattern only blade bodies, or accept duplicates only if they union cleanly.
STL opens at the wrong size	STL units were interpreted incorrectly	Export in millimetres and verify dimensions in the slicer.

19. Recommended beginner workflow

For the first successful build, do not try to optimize everything.

1. Load the **Easy CAD / Easy Print** preset in the simulator.

2. Download the Onshape package.
3. Build only one blade first.
4. Use chord midpoint alignment for every section.
5. Loft the blade.
6. Add a simple hub connector.
7. Export STL.
8. Print one small test if possible.
9. Return to the simulator and improve one variable at a time.

20. What the simulator does not do yet

The current project does not automatically create:

- a finished Onshape document;
- a complete parametric CAD model;
- a print-ready STL;
- supports, slicer settings, or infill settings;
- structural safety verification.

The export is intended to reduce manual copying errors and make CAD rebuilding more consistent. Real wind-tunnel testing is still required for final calibration.

References

- Onshape Help: Insert DXF or DWG into a sketch - <https://cad.onshape.com/help/Content/Sketch/insert_dwg_or_dxf.htm>
- Onshape Help: Loft - <https://cad.onshape.com/help/Content/PartStudio/loft.htm>